Activity: measuring AI speedup

Group members:

AI speedup?

To investigate the effect of AI on user productivity, researchers conducted a randomized experiment with 16 experienced software developers. The authors specified 246 different software development tasks, and randomly assigned each task to be completed with or without AI. The tasks were then divided between the different developers, and the researchers recorded the time it took to complete each task.

Each row in the data represents one task. We will work with the following variables:

- `dev_id`: unique ID for each developer
- `predicted_time_no_ai`: How long the developer estimated the issue would take *without* AI (minutes)
- `ai_treatment`: 0 if AI use was not allowed, 1 if AI use was allowed
- `initial_implementation_time`: Time taken by the developer to solve the issue and create an initial pull request (an initial solution) (minutes)

Questions

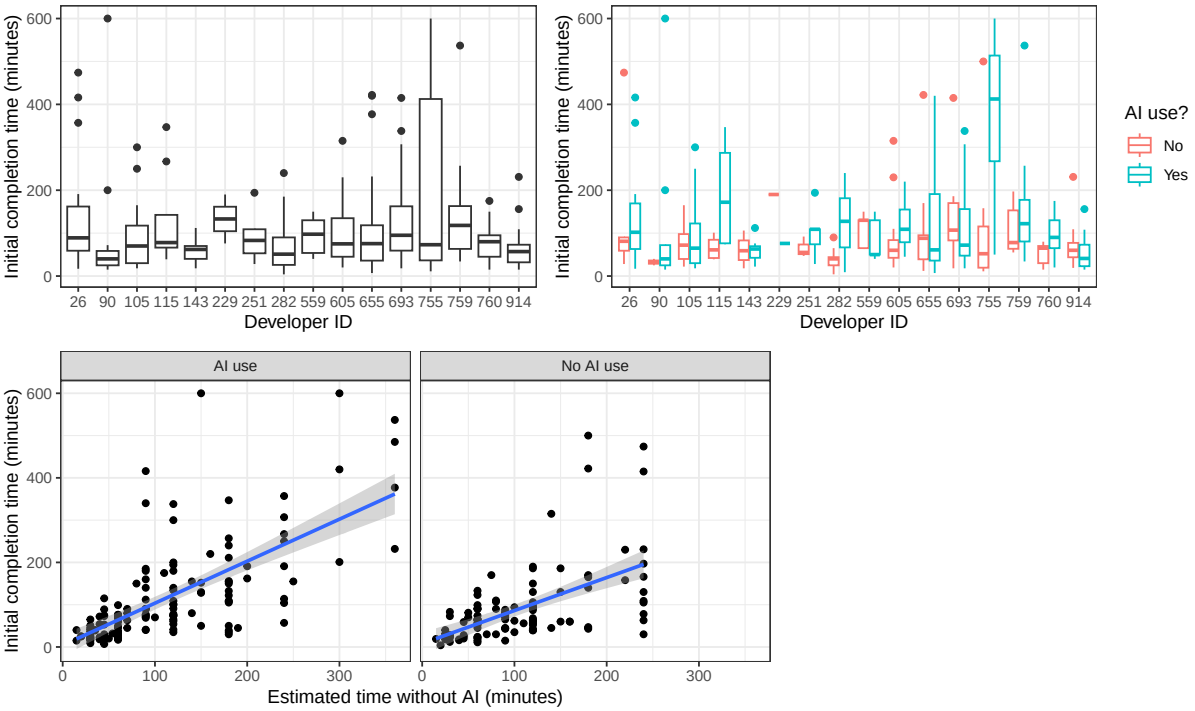
1. Attached is exploratory data analysis for this data and research question. Based on the EDA, write down a regression model that will allow you to address the research question.

Summary of variables

Characteristic	N = 246[1]
dev_id	
26	17 (6.9%)
90	11 (4.5%)
105	19 (7.7%)
115	8 (3.3%)
143	17 (6.9%)
229	2 (0.8%)
251	8 (3.3%)
282	17 (6.9%)
559	10 (4.1%)
605	20 (8.1%)
655	24 (9.8%)
693	28 (11%)
755	11 (4.5%)
759	18 (7.3%)
760	15 (6.1%)
914	21 (8.5%)
predicted_time_no_ai	90 (60, 150)
ai_treatment	
0	110 (45%)
1	136 (55%)
initial_implementation_time	73 (40, 130)

[1] n (%); Median (Q1, Q3)

Plots


Initial completion time (minutes)
600
400
200
0
26 90 105 115 143 229 251 282 559 605 655 693 755 759 760 914
Developer ID
AI use?
No
Yes
AI use
No AI use
Estimated time without AI (minutes)
0 100 200 300